Assignment 4

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1. List all **meta characters used in the Linux shell** and explain their purpose with examples.

Meta Character	Purpose	Example
*	Matches zero or more characters	ls *.txt lists all .txt files
?	Matches exactly one character	ls file?.log matches file1.log, fileA.log
[ ]	Matches any one character within brackets	ls [abc]* matches files starting with a, b, or c
[a-z]	Matches any character in the specified range	ls [0-9]*.txt matches files starting with a digit
[^...] or !	Negates the character set	ls [^a-z]* matches files not starting with lowercase letters
>	Redirects stdout to a file (overwrite)	ls > output.txt saves output to file
>>	Redirects stdout to a file (append)	date >> log.txt appends current date
<	Redirects input from a file	sort < input.txt sorts content of file
&	Runs command in background	sleep 60 & runs sleep in background
&&	Logical AND: runs second command only if first succeeds	mkdir test && cd test
||	Logical OR: runs second command only if first fails	command1 || echo "Failed"
;	Separates multiple commands on same line	cmd1 ; cmd2 runs both regardless of outcome
$()	Command substitution	echo "Today is $(date)"
$	Variable expansion	echo $HOME prints home directory path
\	Escapes special meaning of next character	echo \$HOME prints literal "$HOME"

2. Write a command that uses the **wildcard `*`** to list all `.txt` files in a directory.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls *.txt
text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

3. Write a command that uses the **wildcard `?`** to list files that have exactly five characters in their name.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls
text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls ?????.*
ls: cannot access '?????.*': No such file or directory
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls ??????.*
text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

4. Demonstrate **output redirection** by sending the output of `ls` into a file named `filelist.txt`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls > filelist.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ cat filelist.txt
filelist.txt
text_1.txt
text_2.txt
text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

5. Write a command that **appends output** of the `date` command to a file called `log.txt`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ date
Friday 06 March 2026 11:25:29 PM IST
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ date >> log.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ cat log.txt
Friday 06 March 2026 11:25:37 PM IST
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

6. Show how to **pipe (`|`) the output of `ls` into the `grep` command** to find files containing the word `test`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls | grep 'text'
text_1.txt
text_2.txt
text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

7. Write a command that uses **command substitution `$()`** to display the current date inside an echo statement.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ echo "Current date $(date)"
Current date Friday 06 March 2026 11:28:12 PM IST
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

8. Demonstrate how to use **logical operator `&&`** to run two commands only if the first command succeeds.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ cd ..
digvijay@digvijay-virtualbox:~/Assignment$ cd Assignment_4 && touch sample.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls
filelist.txt  log.txt  sample.txt  text_1.txt  text_2.txt  text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

9. Write a command that runs `sleep 60` in the **background**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ sleep 60 &
[1] 1985
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ps
    PID TTY          TIME CMD
   1821 pts/1    00:00:00 bash
   1985 pts/1    00:00:00 sleep
   1986 pts/1    00:00:00 ps
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

10. Show the command used to **list all background jobs** currently running in the shell.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ jobs
[1]+  Running                 sleep 60 &
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

11. Write a command to **bring a background job to the foreground**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ sleep 60 &
[1] 2021
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ jobs
[1]+  Running                 sleep 60 &
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ fg
sleep 60
```

12. Display the **last 10 commands executed in the shell** using the history command.

```
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ history 10
  333  clear
  334  jobs
  335  jobsclear
  336  clear
  337  jobs
  338  clear
  339  sleep 60 &
  340  jobs
  341  fg
  342  history 10
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

13. Write a command to **execute the last command again** using shell history.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ history 10
  335  jobsclear
  336  clear
  337  jobs
  338  clear
  339  sleep 60 &
  340  jobs
  341  fg
  342  history 10
  343  clear
  344  history 10
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ !!
history 10
  335  jobsclear
  336  clear
  337  jobs
  338  clear
  339  sleep 60 &
  340  jobs
  341  fg
  342  history 10
  343  clear
  344  history 10
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

14. Execute the **5th command from command history** using the appropriate syntax.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ history 10
  337  jobs
  338  clear
  339  sleep 60 &
  340  jobs
  341  fg
  342  history 10
  343  clear
  344  history 10
  345  clear
  346  history 10
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ !346
history 10
  337  jobs
  338  clear
  339  sleep 60 &
  340  jobs
  341  fg
  342  history 10
  343  clear
  344  history 10
  345  clear
  346  history 10
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

15. Create a file named **example.txt** and change its ownership to user `user1`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ touch cdac.txt && sudo chmod 750 cdac.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls -l cdac.txt
-rwxr-x--- 1 digvijay digvijay 0 Mar  6 23:37 cdac.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

16. Write a command to change the **owner and group** of `example.txt` to `user1` and `group1`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ touch example.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls -l example.txt
-rw-rw-r-- 1 digvijay digvijay 0 Mar  6 23:36 example.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ sudo chown user1:group1 example.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls -l example.txt
-rw-rw-r-- 1 user1 group1 0 Mar  6 23:36 example.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

17. Create a file named **cdac.txt** and assign permissions such that the **owner has rwx, group has rx, and others have no permissions**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ touch cdac.txt && sudo chmod 750 cdac.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ ls -l cdac.txt
-rwxr-x--- 1 digvijay digvijay 0 Mar  6 23:37 cdac.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

18. Write a command to **view the Access Control List (ACL)** of a file named `file1`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ touch file1
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ getfacl file1
# file: file1
# owner: digvijay
# group: digvijay
user::rw-
group::rw-
other::r--

digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

19. Write a command to **give user `user2` read and write permissions** on `file1` using ACL.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ setfacl -m u:user1:rw file1
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ getfacl file1
# file: file1
# owner: digvijay
# group: digvijay
user::rw-
user:user1:rw-
group::rw-
mask::rw-
other::r--

digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

20. Write a command to **remove ACL permissions for user `user2`** from `file1`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ setfacl -x u:user1 file1
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ getfacl file1
# file: file1
# owner: digvijay
# group: digvijay
user::rw-
group::rw-
mask::rw-
other::r--

digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

21. Write a command to **mount a filesystem from `/dev/sdb1` to `/mnt/external`**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4 ×
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ sudo mount /dev/sdb1 /mnt/external
mount: /mnt/external: mount point does not exist.
       dmesg(1) may have more information after failed mount system call.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

22. Write a command to **unmount a mounted filesystem from `/mnt/external`**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4 ×
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ sudo umount /mnt/external
umount: /mnt/external: no mount point specified.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

23. Use `grep` to **search for the word "linux" in file.txt ignoring case sensitivity**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4 ×
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ grep -i 'linux' file1
linux
Linux
Linux123
linux1234
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

24. Write a `grep` command to **count how many times the word "error" appears in a file**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4 ×
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ cat file1
error
linux
Linux
Linux123
linux1234
error123
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ grep -c 'error' file1
2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```

25. Write a `find` command to **locate all `.txt` files in the current directory and its subdirectories**.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_4 ×
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$ find . -name '*.txt'
./cdac.txt
./sample.txt
./text_2.txt
./example.txt
./filelist.txt
./text_1.txt
./log.txt
./text_3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_4$
```